

LESSON 5-7 • SESSION 2 • PREREQUISITE PRACTICE B • SKILL 1 OF 2

Surface Area of Rectangular Prisms

Builds on Lesson 5-7 Session 1 • 6.G.A.4

TODAY'S LESSON

You will add all the faces of a rectangular prism. It is the same plan as a box.

YOU NEED TOFind a box's surface area: 3 faces, add, then double. **You will use it in Practice Form A, problems 2 and 8.****STEPS TO FOLLOW***use these same steps on every problem*

- 1 Find** three faces: $l \times w$, $l \times h$, $w \times h$.
- 2 Add** the three face areas.
- 3 Double** the sum. Write square units.

WATCH ME*the steps in action***A box is 6 cm by 4 cm by 3 cm. Find its surface area.****Step 1** Faces: $6 \times 4 = 24$, $6 \times 3 = 18$, $4 \times 3 = 12$.**Step 2** $24 + 18 + 12 = 54$.**Step 3** $2 \times 54 = 108$.**Answer 108 cm²****TRY IT WITH HELP***fill in each step*

1. A box is 5 m by 4 m by 2 m. Find its surface area.

Step 1 Faces: $5 \times 4 =$ _____, $5 \times 2 =$ _____, $4 \times 2 =$ _____**Step 2** $20 + 10 + 8 =$ _____**Step 3** $2 \times 38 =$ _____ m²

2. A box is 10 in by 3 in by 2 in. Find its surface area.

Step 1 Faces: $10 \times 3 =$ _____, $10 \times 2 =$ _____, $3 \times 2 =$ _____**Step 2** $30 + 20 + 6 =$ _____**Step 3** $2 \times 56 =$ _____ in²**YOUR TURN***use the steps on your own*

3. Find the surface area of each box.

a. $3 \times 2 \times 1$ cm _____**b.** $4 \times 4 \times 4$ cm _____**c.** $8 \times 5 \times 4$ cm _____**d.** $6 \times 5 \times 2$ cm _____

Area of Triangles

Builds on Lesson 5-2 Session 1 • 6.G.A.1

TODAY'S LESSONYou will find the triangle ends of a prism, like $\frac{1}{2} \times 24 \times 5$.**YOU NEED TO**Find a triangle's area with its height, not a slanted side. **You will use it in Practice Form A, problems 1, 2, and 7.****STEPS TO FOLLOW***use these same steps on every problem*

- 1 Pick** the base and the height that meet at a square corner.
- 2 Halve** the even number.
- 3 Multiply.** Write square units.

WATCH ME*the steps in action***A right triangle has legs 16 cm and 12 cm. Its long side is 20 cm. Find its area.****Step 1** Base 16 and height 12. Skip the 20.**Step 2** Half of 16 is 8.**Step 3** $8 \times 12 = 96$.**Answer** **96 cm²****TRY IT WITH HELP***fill in each step***4.** Base 12 ft, height 8 ft, slanted sides 10 ft. Find the area.**Step 1** Base _____ and height _____**Step 2** Half of 12 is _____**Step 3** $6 \times 8 =$ _____ ft²**5.** Legs 40 m and 9 m, long side 41 m. Find the area.**Step 1** Base _____ and height _____**Step 2** Half of 40 is _____**Step 3** $20 \times 9 =$ _____ m²**YOUR TURN***use the steps on your own***6.** Find each triangle's area.**a.** legs 6 and 8, long side 10 _____**b.** legs 12 and 35, long side 37 _____**c.** legs 10 and 24, long side 26 _____**d.** base 30, height 8, slanted sides 17 _____**READY CHECK***a problem like the one in today's lesson***7.** Each triangle end has $b = 10$ cm and $h = 6$ cm. What is the area of both ends?**A** 30 cm²**B** 60 cm²**C** 120 cm²**D** 16 cm²**CHECK YOURSELF**
☐ I can do this on my own
 ☐ I need a little help
 ☐ I have a question about #_____